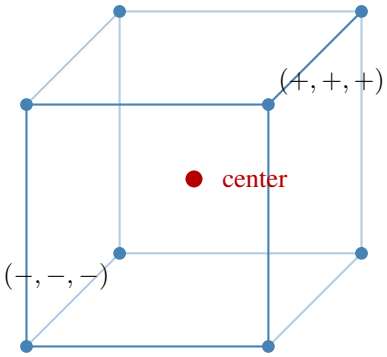
Design of Experiments

A Practical Guide

From Theory to Automated Analysis



First Edition

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Design of Experiments: A Practical Guide

First Edition, 2025

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Preface

This book teaches the theory and practice of Design of Experiments (DOE)—a systematic method for planning experiments that extracts the maximum information from the minimum number of runs. It is written for engineers, scientists, and anyone who runs experiments, benchmarks, or tests and wants to do so more efficiently.

DOE is not new. Ronald Fisher developed the foundations in the 1920s and 1930s for agricultural experiments at Rothamsted. George Box, Genichi Taguchi, and others extended it to manufacturing, quality engineering, and process optimization. Yet the technique remains underused, especially in software engineering and technology, where practitioners often default to changing one variable at a time or running exhaustive grid searches.

This guide takes a practical approach. Each chapter introduces the theory you need, then shows you how to apply it. Mathematical derivations are included where they build intuition, but the emphasis is on understanding *when* and *why* to use each technique, not on proofs.

The examples throughout this book can be run using the open-source DOE Helper Tool that accompanies it. The tool reads a JSON configuration file, generates an optimized experimental design, produces an executable runner script, and analyzes the results with statistical summaries, plots, and reports.

Who this book is for:

- Engineers optimizing systems, processes, or software configurations
- Scientists planning laboratory or field experiments
- Data scientists evaluating model hyperparameters
- Quality engineers improving manufacturing processes
- Anyone who has ever wondered: "Which of these 10 knobs actually matters?"

Prerequisites: Basic familiarity with mean, variance, and standard deviation. No prior statistics coursework is assumed.

Part I

Foundations

Chapter 1

Why Design of Experiments?

1.1 The Cost of Unplanned Experiments

Every experiment costs something—time, money, materials, compute cycles. A poorly planned experiment wastes these resources and may lead to wrong conclusions. A well-designed experiment extracts the maximum learning from each run.

Consider a team tuning a database. They have five configuration knobs they suspect affect performance: buffer pool size, I/O capacity, connection limit, query cache, and thread pool size. The one-variable-at-a-time (OVAT) approach would be:

1. Fix all knobs at their defaults.
2. Change buffer pool size. Measure.
3. Reset buffer pool. Change I/O capacity. Measure.
4. Continue for each knob.

This requires at least 10 runs (2 per knob) and only measures each knob's effect *in isolation, at the default settings of every other knob*. It tells you nothing about how the knobs interact.

A Plackett–Burman screening design tests all five knobs in just 8 runs and estimates every main effect simultaneously. A full factorial on the two most important knobs then takes 4 more runs and reveals their interaction. Twelve runs total, and you know more than OVAT would teach you in fifty.

1.2 The Three Principles of DOE

Fisher established three principles that remain the foundation of every well-designed experiment:

1.2.1 Randomization

Randomization

Randomly assign the order in which experimental runs are performed to protect against unknown lurking variables.

If you run all low-temperature experiments in the morning and all high-temperature experiments in the afternoon, any morning-to-afternoon drift (ambient temperature, operator alertness, equipment warm-up) becomes confounded with your temperature factor. You cannot tell whether the observed difference is due to temperature or time of day.

Randomization does not eliminate lurking variables—it distributes their effects equally across all treatments, on average. This ensures that the statistical tests remain valid.

Mathematically, if Y_{ij} is the response for factor level i in run order j, and there is a time trend δ_j, then without randomization:

$$E[Y_{ij}] = \mu + \tau_i + \delta_j$$

where τ_i and δ_j may be confounded. With randomization, δ_j is equally likely to affect any treatment, so:

$$E[\bar{Y}_{i\cdot}] = \mu + \tau_i$$

and the time trend averages out.

1.2.2 Replication

Replication

Repeat experimental runs to estimate experimental error and increase the precision of effect estimates.

A single run at each factor combination tells you the response at that combination, but it does not tell you how much that response varies. Without an estimate of variability, you cannot determine whether an observed difference is real or just noise.

Replication provides:

- An estimate of the experimental error variance σ^2
- More precise estimates of effects (standard error decreases as $1/\sqrt{n}$)

- The ability to construct confidence intervals and perform significance tests

Replication vs. Repetition

Replication means independently setting up and running the experiment again. **Repetition** means measuring the same setup multiple times. Repetition estimates measurement error, but replication estimates the full experimental error, including setup variation. DOE requires replication.

1.2.3 Blocking

Blocking

Group experimental runs into blocks of similar conditions to remove known sources of variation from the experimental error.

If you must run experiments over two days, make each day a block. Within each block (day), randomize the run order. The day-to-day variation is then accounted for in the analysis and does not inflate the error estimate.

The model becomes:

$$Y_{ijk} = \mu + \tau_i + \beta_j + \epsilon_{ijk}$$

where β_j is the block effect. Because blocks are orthogonal to treatments in a balanced design, the treatment effects τ_i are estimated without bias.

1.3 The OVAT Trap: A Quantitative Example

Suppose the true relationship between yield and two factors (temperature T and pressure P) is:

$$Y = 50 + 10T + 5P + 8TP + \epsilon$$

where $T, P \in \{-1, +1\}$ (coded levels) and $\epsilon \sim N(0, 1)$.

1.3.1 OVAT Analysis

1. Run at $T = -1, P = -1$: $Y = 50 - 10 - 5 + 8 = 43$
2. Change T to $+1$: $Y = 50 + 10 - 5 - 8 = 47$

3. Conclude: Temperature effect $= 47 - 43 = 4$

But the true main effect of temperature is $+10$, not $+4$! The OVAT estimate is biased because it was measured at $P = -1$, where the interaction TP reduces temperature's apparent effect. At $P = +1$, the temperature effect would appear as $+18$.

1.3.2 Factorial Analysis

A 2^2 factorial runs all four combinations:

Run	T	P	Y (expected)
1	-1	-1	43
2	$+1$	-1	47
3	-1	$+1$	37
4	$+1$	$+1$	73

Main effect of T:

$$\text{ME}_T = \frac{(Y_2 + Y_4)}{2} - \frac{(Y_1 + Y_3)}{2} = \frac{47 + 73}{2} - \frac{43 + 37}{2} = 60 - 40 = 20$$

Wait—the formula gives 20, but we said the coefficient was 10. In the effects model, the main effect is *twice* the coefficient because we move from -1 to $+1$ (a change of 2 units). So $\text{ME}_T/2 = 10$, exactly the true coefficient.

The interaction effect:

$$\text{IE}_{TP} = \frac{(Y_1 + Y_4)}{2} - \frac{(Y_2 + Y_3)}{2} = \frac{43 + 73}{2} - \frac{47 + 37}{2} = 58 - 42 = 16$$

Again, $\text{IE}_{TP}/2 = 8$, the true interaction coefficient. The factorial design correctly estimates both effects and their interaction, using the same number of runs where each piece of data contributes to every estimate.

Hidden Replication in Factorials

In a 2^k factorial, every effect estimate uses *all* 2^k observations. Each observation plays a role in estimating every effect. This is why factorials are so efficient—there is no wasted data.

Chapter 2

Statistical Foundations

2.1 The Linear Model

At the heart of DOE analysis is the general linear model. For a single response Y with k factors:

$$Y = \beta_0 + \sum_{i=1}^{k} \beta_i x_i + \sum_{i<j} \beta_{ij} x_i x_j + \epsilon$$

where:

- β_0 is the grand mean (intercept)
- β_i are the main effect coefficients
- β_{ij} are the interaction coefficients
- $x_i \in \{-1, +1\}$ are the coded factor levels
- $\epsilon \sim N(0, \sigma^2)$ is the random error

For a quadratic (response surface) model, add squared terms:

$$Y = \beta_0 + \sum_{i=1}^{k} \beta_i x_i + \sum_{i<j} \beta_{ij} x_i x_j + \sum_{i=1}^{k} \beta_{ii} x_i^2 + \epsilon$$

2.2 Coded Variables

Factor levels are coded to simplify calculations and make effects directly comparable. For a continuous factor with low value L and high value H:

$$x = \frac{\text{actual value} - \frac{L+H}{2}}{\frac{H-L}{2}}$$

This maps $L \to -1$ and $H \to +1$. The center is at $x = 0$. All factors are on the same scale, so their coefficients are directly comparable—a coefficient of 10 means the same magnitude of effect regardless of whether the factor is measured in degrees or bars.

2.3 Estimating Effects

2.3.1 Main Effects

The main effect of factor A is the difference in the average response when A is at its high level versus its low level:

$$\text{ME}_A = \bar{Y}_{A+} - \bar{Y}_{A-}$$

In a 2^k design, each average uses half the total runs, so the standard error of a main effect is:

$$\text{SE}(\text{ME}_A) = \sqrt{\frac{4s^2}{N}} = \frac{2s}{\sqrt{N}}$$

where s^2 is the pooled variance estimate and N is the total number of runs.

2.3.2 Interaction Effects

The two-factor interaction AB measures whether the effect of A depends on the level of B. Compute it as:

$$\text{IE}_{AB} = \frac{1}{2}\big[(\text{effect of } A \text{ at } B_+) - (\text{effect of } A \text{ at } B_-)\big]$$

Equivalently, in a 2^k design:

$$\text{IE}_{AB} = \bar{Y}_{\text{concordant}} - \bar{Y}_{\text{discordant}}$$

where "concordant" runs have A and B both high or both low, and "discordant" runs have one high and one low.

2.3.3 Percentage Contribution

To rank factors by importance, compute each factor's percentage contribution:

$$\text{PC}_i = \frac{|\text{ME}_i|}{\sum_{j=1}^{k} |\text{ME}_j|} \times 100\%$$

This appears on the Pareto chart. The "vital few" factors typically account for 70–80% of the total effect magnitude.

2.4 Confidence Intervals

A 95% confidence interval for a main effect is:

$$\text{ME}_A \pm t_{\alpha/2,\nu} \cdot \text{SE}(\text{ME}_A)$$

where $t_{\alpha/2,\nu}$ is the critical value from the t-distribution with ν degrees of freedom. If the interval contains zero, the effect is not statistically significant at the 5% level.

For a two-level design with n_- and n_+ observations at the low and high levels respectively:

$$\text{SE}(\text{ME}_A) = s_p \sqrt{\frac{1}{n_-} + \frac{1}{n_+}}$$

where s_p is the pooled standard deviation:

$$s_p = \sqrt{\frac{(n_- - 1)s_-^2 + (n_+ - 1)s_+^2}{n_- + n_+ - 2}}$$

2.5 Analysis of Variance (ANOVA)

ANOVA decomposes the total variation in the response into components attributable to each factor, their interactions, and the residual error:

$$\text{SS}_{\text{Total}} = \sum_i \text{SS}_{A_i} + \sum_{i<j} \text{SS}_{A_i A_j} + \text{SS}_{\text{Error}}$$

For a main effect in a 2^k design:

$$\text{SS}_A = \frac{N}{4} \cdot \text{ME}_A^2$$

The F-statistic for testing whether factor A has a significant effect is:

$$F_A = \frac{\text{SS}_A/1}{\text{SS}_{\text{Error}}/\nu_E} = \frac{\text{MS}_A}{\text{MS}_E}$$

If $F_A > F_{\alpha,1,\nu_E}$, reject the null hypothesis that factor A has no effect.

2.6 The R^2 Statistic

R^2 measures the proportion of variance in the response explained by the model:

$$R^2 = 1 - \frac{\text{SS}_{\text{Error}}}{\text{SS}_{\text{Total}}}$$

The adjusted R^2 penalizes for model complexity:

$$R^2_{\text{adj}} = 1 - \frac{\text{SS}_{\text{Error}}/\nu_E}{\text{SS}_{\text{Total}}/\nu_T}$$

An R^2 near 1.0 means the model explains nearly all the variation. An R^2 below 0.7 suggests important factors or relationships are missing from the model.

Part II

Design Types

Chapter 3

Full Factorial Designs

3.1 The 2^k Factorial

The 2^k full factorial design tests every combination of k factors at 2 levels each. It is the most complete two-level design: it estimates all main effects, all two-factor interactions, and all higher-order interactions with no confounding.

3.1.1 Run Count

Factors (k)	Runs (2^k)	Notes
2	4	Minimum useful design
3	8	Very practical
4	16	Still manageable
5	32	Getting expensive
6	64	Consider fractional
7	128	Fractional almost always better

3.1.2 The Design Matrix

For a 2^3 design with factors A, B, C:

Run	A	B	C	AB	AC	BC	ABC
1	$-$	$-$	$-$	$+$	$+$	$+$	$-$
2	$+$	$-$	$-$	$-$	$-$	$+$	$+$
3	$-$	$+$	$-$	$-$	$+$	$-$	$+$
4	$+$	$+$	$-$	$+$	$-$	$-$	$-$
5	$-$	$-$	$+$	$+$	$-$	$-$	$+$
6	$+$	$-$	$+$	$-$	$+$	$-$	$-$
7	$-$	$+$	$+$	$-$	$-$	$+$	$-$
8	$+$	$+$	$+$	$+$	$+$	$+$	$+$

The interaction columns are computed by multiplying the corresponding main effect columns element-wise. This orthogonality property—every column has an equal number of $+$ and $-$ signs, and the inner product of any two columns is zero—is what makes the effect estimates independent and unbiased.

3.1.3 Geometric Interpretation

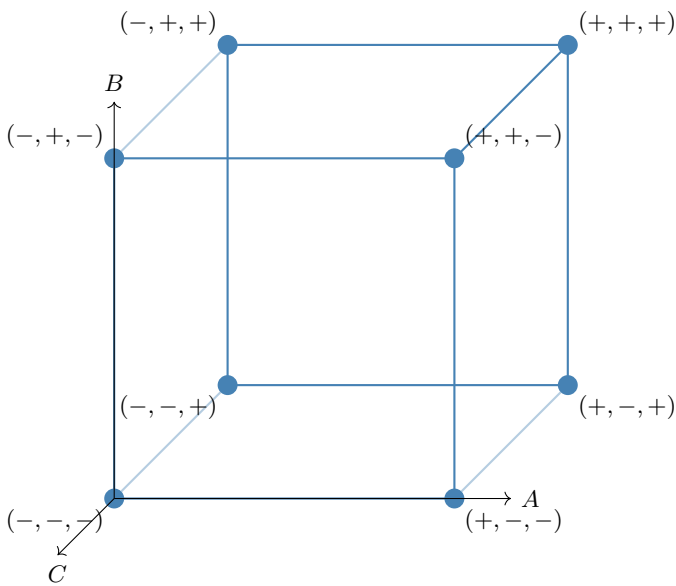


Figure 3.1: A 2^3 full factorial as a cube. Each vertex is an experimental run. The main effect of A is the average of the four right-face responses minus the average of the four left-face responses.

3.2 Mixed-Level Factorials

Full factorials are not limited to two levels. A $3 \times 2 \times 4$ design tests one factor at 3 levels, one at 2, and one at 4, yielding $3 \times 2 \times 4 = 24$ runs. The analysis uses the range of level means (maximum mean minus minimum mean) instead of the simple high-minus-low formula.

3.3 When to Use Full Factorials

- 2–4 factors: almost always the right choice
- You need to estimate all interactions, not just main effects
- Runs are cheap relative to the information gained
- You suspect important interactions exist

Chapter 4

Fractional Factorial Designs

4.1 The Problem of Exponential Growth

As the number of factors increases, full factorials become impractical. With 7 factors at 2 levels, a full factorial requires $2^7 = 128$ runs. But in practice, the *effect sparsity principle* states that most of the variation is explained by main effects and low-order interactions. Higher-order interactions (ABC, $ABCD$, etc.) are usually negligible.

Fractional factorials exploit this by deliberately confounding higher-order interactions with main effects or lower-order interactions, trading information about unlikely-to-matter effects for a dramatic reduction in runs.

4.2 The 2^{k-p} Design

A 2^{k-p} fractional factorial uses 2^{k-p} runs to study k factors. The integer p is the *degree of fractionation*: each increase in p halves the number of runs.

Design	Factors	Runs	Fraction
2^{4-1}	4	8	$\frac{1}{2}$
2^{5-1}	5	16	$\frac{1}{2}$
2^{5-2}	5	8	$\frac{1}{4}$
2^{7-4}	7	8	$\frac{1}{16}$

4.3 Resolution

The *resolution* of a fractional factorial describes the severity of the confounding:

Resolution	Confounding Pattern
III	Main effects aliased with 2-factor interactions
IV	Main effects clear; 2FIs aliased with other 2FIs
V	Main effects and 2FIs clear; 2FIs aliased with 3FIs

Choosing Resolution

Use Resolution III for initial screening (many factors, few runs). Use Resolution IV or V when you suspect important 2-factor interactions and want to estimate them separately from main effects. The higher the resolution, the more runs are required.

4.4 Generator Strings and Aliasing

A fractional factorial is defined by its *generators*—the rules that assign aliased factor columns. For a 2^{5-2}_{III} design, the base factors are A, B, C (requiring $2^3 = 8$ runs), and the two additional factors are defined as:

$$D = AB$$
$$E = AC$$

The *defining relation* is the set of column products equal to I (the identity):

$$I = ABD = ACE = BCDE$$

Any two columns whose product appears in the defining relation are aliased (confounded). For example, $A = BD = CE$ and $B = AD = CDE$.

4.5 Practical Guidance

1. If you have $k \leq 4$ factors, use a full factorial.
2. If $5 \leq k \leq 7$, use a Resolution IV or V fraction.

3. If $k > 7$ and you are screening, use Resolution III or a Plackett–Burman.
4. Always check the alias structure before running the experiment to ensure the effects you care about are not aliased with each other.

Chapter 5

Screening Designs

5.1 Plackett–Burman Designs

When you have many factors (5–20 or more) and your primary goal is to identify the *vital few* that have large main effects, Plackett–Burman (PB) designs are the most efficient choice.

PB designs are 2-level designs where the number of runs is always a multiple of 4. They can screen up to $N - 1$ factors in N runs:

Runs (N)	Max Factors ($N - 1$)
8	7
12	11
16	15
20	19
24	23

5.1.1 Properties

- All main effects are estimated with equal precision.
- Main effects are partially confounded with two-factor interactions (complex aliasing, unlike the clean aliasing of fractional factorials).
- The design is *not* suitable for estimating interactions—it is a pure screening tool.

5.1.2 The Screening-to-Optimization Pipeline

PB designs are almost always the first step in a sequential strategy:

1. **Screen** with PB: Run N experiments, identify the 3–5 factors with the largest main effects.
2. **Characterize** with a full or fractional factorial on the important factors.
3. **Optimize** with a response surface design (CCD or Box–Behnken).

Chapter 6

Response Surface Designs

6.1 Beyond Main Effects

Screening and factorial designs estimate linear effects: how much does the response change when a factor moves from low to high? But many real systems have *curvature*—the optimum is not at an extreme, but somewhere in the middle. To find this optimum, we need a quadratic model:

$$Y = \beta_0 + \sum_i \beta_i x_i + \sum_{i<j} \beta_{ij} x_i x_j + \sum_i \beta_{ii} x_i^2 + \epsilon$$

Response surface designs provide the data points needed to fit this model. They include runs at the center of the factor space (to estimate curvature) and, for CCD, star points beyond the factorial range.

6.2 Central Composite Design (CCD)

The CCD consists of three components:

1. **Factorial points:** A 2^k (or 2^{k-p}) factorial at the corners.
2. **Axial (star) points:** $2k$ points along each axis at distance α from the center.
3. **Center points:** Several runs at the center of the design space.

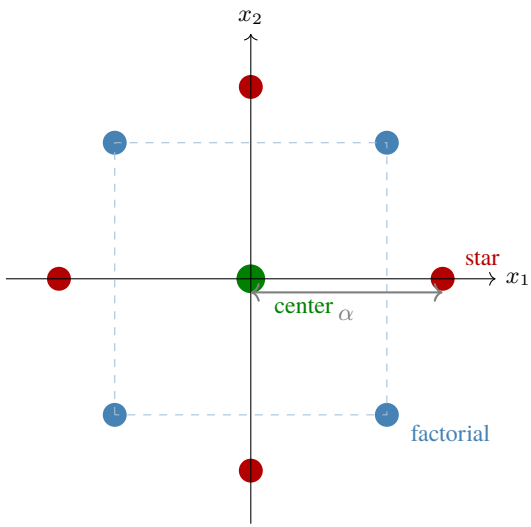


Figure 6.1: Central Composite Design for 2 factors. Blue squares are factorial points, red circles are star (axial) points, and the green circle is the center point.

The value of α determines the design's properties:

- $\alpha = 1$: Face-centered CCD (star points on the face of the cube). Does not require runs outside the factor range, but the quadratic fit is less precise.
- $\alpha = \sqrt{k}$: Spherical CCD. Star points lie on the sphere circumscribing the factorial cube.
- $\alpha = (2^k)^{1/4}$: Rotatable CCD. The prediction variance is constant at all points equidistant from the center, making the model equally precise in all directions.

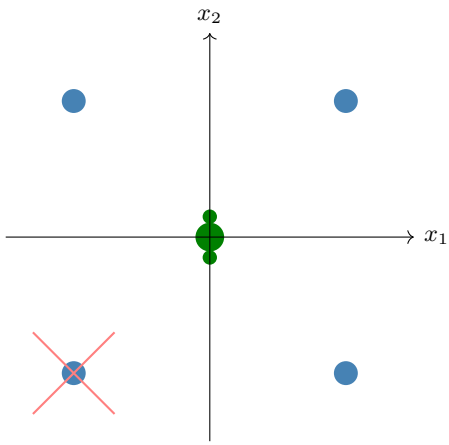
6.2.1 Run Count

Factors	Factorial	Star	Center	Total
2	4	4	5	13
3	8	6	6	20
4	16	8	7	31
5	32	10	8	50

6.3 Box–Behnken Design

Box–Behnken Advantage

Box–Behnken designs avoid extreme corners of the design space. Every run has at most two factors at their extreme levels; the remaining factors are at the center. This makes them safer for physical experiments where corner points are dangerous or infeasible.



(x_3 at center for each pair)

Figure 6.2: Box–Behnken design concept. The extreme corner (crossed out) is not tested. Each pair of factors is tested at all four edge combinations while the third stays at center.

For 3 factors, Box–Behnken requires 15 runs (12 edge-midpoint runs + 3 center replicates), compared to 20 for a CCD. For 4 factors: 27 runs.

6.3.1 When to Use Box–Behnken vs. CCD

Criterion	CCD	Box–Behnken
Avoids extreme corners	No	Yes
Points outside factor range	Yes (star)	No
Minimum factors	2	3
Sequential augmentation from 2^k	Yes	No
Rotatable option	Yes	No
Fewer runs (3–4 factors)	Depends	Often

6.4 Latin Hypercube Sampling

Latin Hypercube Sampling (LHS) is a space-filling design used primarily for computer experiments and simulations where the response is deterministic (no random error).

6.4.1 Construction

For n samples and k factors:

1. Divide each factor's range into n equal intervals.
2. Place exactly one sample point in each interval of each factor.
3. The placement within each interval is random.

This guarantees that the projection onto any single factor axis covers the entire range uniformly—a property called *stratification*.

6.4.2 When to Use LHS

- Computer experiments with deterministic outputs (no noise)
- High-dimensional spaces (many factors) where factorial designs are infeasible
- Building surrogate models (kriging, Gaussian processes)
- Sensitivity analysis and uncertainty quantification

Part III

Analysis and Optimization

Chapter 7

Analyzing Results

7.1 The Pareto Chart

The Pareto chart is the single most useful visualization in DOE. It displays factors sorted by the absolute magnitude of their main effects, with a cumulative percentage line overlaid.

The 80/20 Rule

The Pareto principle states that roughly 80% of the total effect magnitude is caused by 20% of the factors. The Pareto chart's cumulative line helps you identify these "vital few" factors. Factors below the 80% line are candidates for being fixed at their cheapest or most convenient level.

7.1.1 Reading a Pareto Chart

1. Factors are sorted from largest to smallest absolute effect (top to bottom or left to right).
2. The bar length represents the magnitude of the main effect.
3. The cumulative line shows the running total as a percentage of the sum of all effect magnitudes.
4. The 80% reference line marks where the "vital few" end and the "trivial many" begin.

7.2 Main Effects Plots

A main effects plot shows the mean response at each level of a factor, connected by line segments. The slope of the line indicates the magnitude and direction of the effect.

- A steep line means a large effect.
- A flat line means the factor has little influence.
- A non-linear pattern (for >2 levels) suggests curvature.

7.3 Interaction Plots

An interaction plot shows the mean response for each combination of two factors. If the lines are parallel, there is no interaction. If they cross or diverge, the interaction is significant.

7.4 Statistical Significance

7.4.1 Using Confidence Intervals

For each main effect, compute the 95% confidence interval:

$$\text{CI} = \text{ME} \pm t_{0.025,\nu} \cdot \text{SE}$$

If the interval does not contain zero, the effect is statistically significant. Report the confidence interval alongside the point estimate—it conveys both the magnitude and the uncertainty.

7.4.2 Practical vs. Statistical Significance

A factor can be statistically significant (the CI excludes zero) but practically unimportant (the effect is tiny). Conversely, a factor can be practically important but not statistically significant (the experiment was too small to detect it). Always consider both.

Do Not Over-Rely on p-Values

A p-value tells you how unlikely the observed data would be if the true effect were zero. It does *not* tell you the probability that the effect is real, nor the magnitude of the effect. Focus on effect sizes and confidence intervals.

Chapter 8

Response Surface Methodology

8.1 From Effects to Models

Screening and factorial designs answer the question: "Which factors matter?" Response Surface Methodology (RSM) answers: "What are the optimal settings?"

RSM fits a polynomial model to the data and uses the model to predict the response at untested factor combinations, including the predicted optimum.

8.2 Fitting the Model

Given n observations, the linear model in matrix form is:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

The least-squares estimate is:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

For a linear model with k factors, $\mathbf{X}$ is an $n \times (k+1)$ matrix (including the intercept column). For a quadratic model, $\mathbf{X}$ has $1 + 2k + \binom{k}{2}$ columns (intercept + linear + squared + interaction terms).

8.3 Model Adequacy

8.3.1 R^2 and Adjusted R^2

R^2 always increases when you add terms to the model, even if those terms are pure noise. Adjusted R^2 penalizes for model complexity:

- $R^2 > 0.9$: Excellent fit
- $0.7 < R^2 < 0.9$: Good fit, some unexplained variation
- $R^2 < 0.7$: Consider adding terms or transforming variables

A large gap between R^2 and R^2_{adj} suggests overfitting.

8.3.2 Residual Analysis

Always check the residuals $e_i = y_i - \hat{y}_i$:

- Plot residuals vs. fitted values: should show no pattern
- Plot residuals vs. run order: should show no trend (drift)
- Normal probability plot of residuals: should be approximately linear

8.4 Finding the Optimum

For a quadratic model, the stationary point (where the partial derivatives are zero) can be found analytically:

$$\hat{\mathbf{x}}_s = -\frac{1}{2}\mathbf{B}^{-1}\mathbf{b}$$

where $\mathbf{b}$ is the vector of linear coefficients and $\mathbf{B}$ is the matrix of quadratic and interaction coefficients.

If the stationary point is a maximum (all eigenvalues of $\mathbf{B}$ are negative), it is the predicted optimum for a "maximize" response. If it is a minimum (all eigenvalues positive), it is the optimum for "minimize." If eigenvalues have mixed signs, the stationary point is a saddle point, and the optimum lies on the boundary of the design space.

Confirmation Runs

Never trust a predicted optimum without verification. Run 3–5 confirmation experiments at the predicted settings. If the observed response matches the prediction (within the confidence interval), the model is validated.

8.5 Multi-Response Optimization

When multiple responses must be optimized simultaneously (e.g., maximize yield while minimizing cost), the *desirability function* approach is standard:

1. For each response i, define a desirability d_i that maps the response to a $[0, 1]$ scale (0 = unacceptable, 1 = ideal).
2. Compute the overall desirability as the geometric mean: $D = (d_1 \cdot d_2 \cdot \ldots \cdot d_m)^{1/m}$
3. Find the factor settings that maximize D.

Part IV

Practice

Chapter 9

The Sequential Experimentation Strategy

9.1 Phase 1: Screening

Goal: Identify the vital few factors from a large list of candidates.

Design: Plackett–Burman or Resolution III fractional factorial.

Runs: 8–20 (regardless of how many factors you have).

Analysis: Pareto chart, main effects. Ignore interaction estimates (they are confounded). Rank factors by absolute effect magnitude.

Decision: Select the top 3–5 factors for further study. Fix the rest at their cheapest, easiest, or most robust level.

9.2 Phase 2: Characterization

Goal: Understand main effects *and* interactions among the important factors.

Design: Full factorial (2^k with $k = 3$–5) or Resolution V fractional factorial.

Runs: 8–32.

Analysis: Full ANOVA, interaction effects, confidence intervals. Check for curvature by including center points—if the center-point mean differs significantly from the average of the corner points, curvature exists.

Decision: If no curvature, the optimal settings are at the best corner. If curvature exists, proceed to Phase 3.

9.3 Phase 3: Optimization

Goal: Find the optimal factor settings within the design space.

Design: Central Composite Design or Box–Behnken.

Runs: 15–50 (depending on the number of factors).

Analysis: Fit a quadratic RSM model. Check R^2, residuals, and the location of the stationary point.

Decision: Identify the predicted optimum. If it is within the design space, proceed to confirmation. If it is on the boundary, consider shifting the design space and running a new experiment.

9.4 Phase 4: Confirmation

Goal: Validate the predicted optimum.

Design: 3–5 replicate runs at the predicted optimal settings.

Analysis: Compare the observed mean to the predicted value. If the observed value falls within the prediction interval, the model is confirmed.

Chapter 10

Applications

10.1 Software Performance Benchmarking

Software systems have many tunable parameters—thread pool sizes, buffer capacities, timeout values, caching strategies—and engineers often tune them by intuition, one at a time. DOE provides a systematic alternative.

10.1.1 Example: Database Configuration

A database administrator suspects five configuration knobs affect transaction throughput:

Factor	Low	High
`buffer_pool_size`	1 GB	4 GB
`io_capacity`	200	2000
`max_connections`	100	500
`query_cache`	0 MB	256 MB
`thread_pool_size`	4	16

Phase 1: A Plackett–Burman design with 8 runs reveals that `buffer_pool_size` and `io_capacity` account for 75% of the throughput variation. The other three knobs have negligible effects.

Phase 2: A 2^2 full factorial on the two important knobs (4 runs, with 2 center points) confirms a significant interaction: large buffer pools only help when I/O capacity is also high.

Phase 3: A CCD with 13 runs on the two knobs finds the optimal settings: `buffer_pool_size` = 3.2 GB, `io_capacity` = 1600.

Total: 27 runs. An OVAT approach would have tested each knob at 3–4 values (≈15–20 runs) and missed the interaction entirely.

10.1.2 Example: Compiler Flags

A performance engineer wants to understand how compiler optimization level (`-O1`, `-O2`, `-O3`), link-time optimization (`off`, `thin`, `full`), and architecture target (`native`, `x86-64-v3`) affect runtime and binary size.

A $3 \times 3 \times 2 = 18$ full factorial tests all combinations. The analysis reveals that `-O3` with `thin` LTO produces the best runtime, but `full` LTO produces the smallest binary. The interaction between optimization level and LTO is significant: `full` LTO improves `-O1` more than `-O3` (diminishing returns when the optimizer has already done most of the work).

10.2 Scientific and Engineering Experiments

10.2.1 Example: Chemical Reactor Optimization

A chemical engineer wants to maximize yield and purity while minimizing cost in a batch reactor. Three factors are controlled: temperature (150–200°C), pressure (2–6 bar), and catalyst concentration (0.5–2.0 g/L).

Running the reactor at all extremes simultaneously (200°C, 6 bar, 2.0 g/L) risks a runaway reaction. A Box–Behnken design avoids this: each run has at most two factors at their extremes while the third is at center.

With 15 runs, the engineer fits a quadratic model for each response and discovers that:

- Temperature has the largest effect on yield (quadratic, with a maximum near 185°C).
- Catalyst concentration has the largest effect on purity (linear, higher is better).
- Cost increases linearly with all three factors, with a temperature–pressure interaction.

The desirability function approach finds the compromise: 185°C, 3.5 bar, 1.8 g/L, yielding 78% yield, 95% purity, at $55/batch.

10.2.2 Example: Materials Science

A materials scientist studies the tensile strength of a new alloy as a function of four processing parameters. A CCD with 31 runs reveals that annealing temperature has a strong quadratic effect (optimum at 520°C) and interacts

with cooling rate. The quadratic model achieves $R^2 = 0.94$ and correctly predicts the confirmation runs within $\pm 3\%$.

10.3 Machine Learning Hyperparameter Tuning

DOE can replace random search or grid search for hyperparameter optimization:

- **Screening:** Use a PB design to test 10+ hyperparameters and identify the 3–4 that actually affect validation accuracy.
- **Optimization:** Use a CCD or LHS on the important hyperparameters to find the optimal region.
- **Advantage over grid search:** A 2^{10} grid search requires 1024 training runs. A PB screen requires 12.
- **Advantage over random search:** LHS guarantees better space coverage than pure random sampling with the same number of points.

Chapter 11

Using the DOE Helper Tool

11.1 Installation

```
# Clone the repository
git clone <repository-url>
cd design_of_experiments

# Install dependencies
pip install -r requirements.txt

# Or install as a package
pip install -e .
```

11.2 Configuration File

The tool is driven by a JSON configuration file:

```
{
  "metadata": {
    "name": "My Experiment",
    "description": "What I am investigating"
  },
  "factors": [
    {"name": "temperature", "levels": ["150", "200"],
     "type": "continuous", "unit": "C"},
    {"name": "catalyst", "levels": ["A", "B"],
     "type": "categorical"}
```

```
    ],
    "responses": [
      {"name": "yield", "optimize": "maximize", "unit": "%"},
      {"name": "cost",  "optimize": "minimize", "unit": "USD"}
    ],
    "settings": {
      "operation": "full_factorial",
      "test_script": "my_test.sh",
      "out_directory": "results",
      "block_count": 1
    }
  }
```

11.3 Supported Operations

Operation	Design Type	Requirements
`full_factorial`	All combinations	Any levels
`fractional_factorial`	2^{k-p} fraction	2 levels
`plackett_burman`	PB screening	2 levels
`latin_hypercube`	Space-filling	Any
`central_composite`	CCD	2 numeric levels
`box_behnken`	Box–Behnken	2 numeric, $\geq$3 factors

11.4 Workflow

```
# 1. Preview the design
doe info --config config.json

# 2. Generate runner script
doe generate --config config.json --output run.sh --seed 42

# 3. Execute experiments
bash run.sh

# 4. Analyze results
doe analyze --config config.json

# 5. Get optimization recommendations
doe optimize --config config.json
```

```
# 6. Generate HTML report
doe report --config config.json --output report.html

# 7. Export to CSV
doe analyze --config config.json --csv results/csv
```

11.5 Test Script Protocol

Your test script must:

1. Accept factor values via the configured argument style (`--name value`, environment variables, or positional arguments).
2. Accept `--out <path>` to specify the output file.
3. Write a JSON file to the output path with keys matching your response names.

```
#!/bin/bash
# Example test script
# ... run your experiment here ...
echo '{"yield": 85.3, "cost": 12.50}' > "$OUT_PATH"
```

11.6 Analysis Output

The `analyze` command computes and displays:

- Main effects with standard errors and 95% confidence intervals
- Two-factor interaction effects (for 2-level factors)
- Summary statistics (count, mean, std, min, max) per factor per level
- Pareto chart with cumulative contribution line
- Main effects plots (grid of line plots, one per factor)

The `optimize` command additionally fits a linear RSM model and reports R^2, coefficients, predicted optimum, and factor importance ranking.

The `report` command generates a self-contained HTML file with all tables, plots (embedded as base64 images), and a collapsible design matrix.

Appendix A

Mathematical Reference

A.1 Effect Estimation Formulas

For a 2^k factorial with $N = 2^k$ runs, coded levels $x \in \{-1, +1\}$, and response vector $\mathbf{y}$:

Main effect of factor i:

$$\mathrm{ME}_i = \frac{2}{N} \sum_{j=1}^{N} x_{ij} y_j = \bar{y}_{i+} - \bar{y}_{i-}$$

Interaction effect of factors i and k:

$$\mathrm{IE}_{ik} = \frac{2}{N} \sum_{j=1}^{N} x_{ij} x_{kj} y_j$$

Standard error of any effect:

$$\mathrm{SE} = \frac{2s}{\sqrt{N}}$$

where $s^2 = \mathrm{MS}_E$ (mean square error from ANOVA or from replicates).

Sum of squares for any effect:

$$\mathrm{SS}_{\text{effect}} = \frac{N}{4} \cdot \mathrm{ME}^2$$

A.2 Response Surface Model

For k factors, the full quadratic model has $p = 1 + 2k + \binom{k}{2}$ parameters:

$$\hat{y} = b_0 + \sum_{i=1}^{k} b_i x_i + \sum_{i=1}^{k} b_{ii} x_i^2 + \sum_{i<j} b_{ij} x_i x_j$$

k	p (parameters)
2	6
3	10
4	15
5	21

The design must have at least p distinct runs (and preferably $\geq 1.5p$) to fit the model.

A.3 Minimum Run Counts by Design

	Number of Factors					
Design	2	3	4	5	6	7
Full factorial	4	8	16	32	64	128
Frac. factorial (Res. III)	–	4	8	8	8	8
Plackett–Burman	4	4	8	8	8	8
CCD	13	20	31	52	–	–
Box–Behnken	–	15	27	46	–	–

Appendix B

Glossary

ANOVA Analysis of Variance. Partitions total variation into components.

Blocking Grouping runs to account for known nuisance variation.

CCD Central Composite Design. A response surface design with factorial, star, and center points.

Coded variable A factor rescaled so that -1 = low and $+1$ = high.

Confounding When two effects cannot be distinguished because they produce identical patterns in the design matrix.

DOE Design of Experiments.

Effect The change in the mean response caused by changing a factor from one level to another.

Factor A controllable input variable in an experiment.

Fractional factorial A subset of a full factorial that trades higher-order information for fewer runs.

Interaction When the effect of one factor depends on the level of another.

Level A specific value or setting of a factor.

LHS Latin Hypercube Sampling. A space-filling design.

Main effect The average effect of a factor across all levels of other factors.

OVAT One Variable At a Time. An inefficient experimental strategy.

Pareto chart A bar chart ranking effects by magnitude.

Plackett–Burman An efficient screening design for identifying important main effects.

Randomization Randomly ordering experimental runs to protect against lurking variables.

Replication Independently repeating experimental runs.

Resolution A measure of the confounding severity in a fractional factorial.

Response A measurable outcome of an experiment.

RSM Response Surface Methodology. Fitting polynomial models to find optimal settings.

R^2 Coefficient of determination. Proportion of variance explained by the model.

References

[1] Box, G.E.P., Hunter, J.S., & Hunter, W.G. (2005). *Statistics for Experimenters: Design, Innovation, and Discovery*, 2nd ed. Wiley.

[2] Montgomery, D.C. (2017). *Design and Analysis of Experiments*, 9th ed. Wiley.

[3] Myers, R.H., Montgomery, D.C., & Anderson-Cook, C.M. (2016). *Response Surface Methodology: Process and Product Optimization Using Designed Experiments*, 4th ed. Wiley.

[4] Box, G.E.P. & Behnken, D.W. (1960). Some new three level designs for the study of quantitative variables. *Technometrics*, 2(4), 455–475.

[5] Plackett, R.L. & Burman, J.P. (1946). The design of optimum multifactorial experiments. *Biometrika*, 33(4), 305–325.

[6] Fisher, R.A. (1935). *The Design of Experiments*. Oliver & Boyd.

[7] NIST/SEMATECH e-Handbook of Statistical Methods. Available at: https://www.itl.nist.gov/div898/handbook/